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A Robust Multi-Channel EEG Signals Preprocessing Method for Enhanced Upper Extremity Motor Imagery Decoding

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***Abstract*- Brain computer interface (BCI) based on non-invasive electroencephalography (EEG) signals has become a promising and alternative method for electromyography (EMG) signal in the rehabilitation process of amputees with high level of amputation. This is because their residual muscles cannot provide sufficient myoelectric signals to accurately recognize limb movement intentions. One of the major challenges of BCI based EEG methods is the inevitable artifacts contained in multi-channels EEG recordings that would affect accurate characterization and decoding of limb movement intents from brain signals. Previous studies have applied different methods based on regression, blind source separation (BSS) and filtering techniques, though with limited efficacy in fully isolating artifacts from the entire EEG signals. Also, some of the existing methods are targeted towards the removal of one or two particular types of artifacts. In this study, a linear algorithm based on generalized eigenvalue decomposition and multi-channel Wiener filter (GEVD-MWF) was proposed to simultaneously remove different type of artifacts inherent in EEG signals, recorded from four transhumeral amputee subjects. Experimental results showed that the proposed method outperformed the commonly used approach which achieved an improved average classification accuracy of 22.03% across all the classes of motor imagery (MI) tasks and subjects. For real-time applications, a modified channel selection algorithm-based on sequential forward floating selection (mSFFS) was adopted to remove redundant channels and an average accuracy of 96.87% was achieved with 10 electrode channels located at the pre-motor cortex, primary motor cortex and somatosensory cortex of the brain. The overall performance of the proposed method suggested that multiclass MI tasks can be reliably and accurately characterized using artifact-free EEG signals from a few number of channels. Thus, the proposed methods can be a potential control method for rehabilitation devices for individuals with high level amputation.**

Index terms:* *Amputation, Artifact removal, Electroencephalography signals, Classifiers, Multi-channel Wiener filter.

I. INTRODUCTION

The use of electromyogram based pattern recognition (EMG -PR) method to drive upper limb prostheses has received considerable attention in the last decades. This method provides a more intuitive control with multiple degrees of freedom capabilities, thereby bringing the prosthesis closer to the natural arm function. Similarly, due to the rich neural information set inherent in myoelectric signals, the intention of individual limb movements performed by persons with upper amputation could be accurately decoded [1]. Though, this method has many advantages, but its practical use in controlling prostheses for people with high degree of amputation (such as upper elbow and shoulder disarticulation amputees) is still a major challenge. This is because the residual arm muscles of these individuals may either not be available (shoulder disarticulation) or insufficient (upper elbow amputation) to generate enough myoelectric signals to identify limb motion intentions. Therefore, the commercial availability of multifunctional prostheses for these group of individuals are scarce [2-4].

Alternative methods adopted in the time past to address this issue include: brain computer interface (BCI), peripheral nerve interface (PNI) and targeted muscle innervation (TMR) [5]. Comparing the above mentioned methods, BCI based on electroencephalogram (EEG) signal showed a great promise for individuals with high level amputation due to its non-invasiveness, ease of use, portability, good temporal resolution and cost effectiveness [6-7]. However, EEG signals are susceptible to various non-physiological (also known as extrinsic/external artifact which occur as a result of electrode displacement, electromagnetic interference, etc.) and physiological artifacts. Through the use of simple filter, proper planning and approaches, the effects of non-neurophysiological artifacts can be minimized or avoided. However, physiological artifacts caused by eye movements and blinks, cardiac activity, head movements and muscle activity are more difficult to

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eliminate because they require the use of artifacts removal algorithms [8-9].

Previous studies have successfully explored different methods based on regression, blind source separation (BSS) techniques to removal various types of artifacts from multichannel EEG signals. For instance, Wang et al. [10] introduced an automatic artifact correction method based on regression and independent component analysis (ICA) to recover the original source signals for multi-class motor imagery (MI) EEG decoding. De Vos et al. [11] applied Canonical Correlation Analysis (CCA) to remove muscle artifacts. A novel discriminative ocular artifact correction approach (oscillatory correlations) for feature learning in EEG analysis was proposed in [12]. Furthermore, a comparative study was conducted on automatic artifact removal (AAR), fully automated statistical thresholding for EEG artifact Rejection (FASTER) and multiple artifact rejection algorithm (MARA) for robust classification of motor imagery tasks [13].

Generally, some of these methods are semi-automatic while other are fully automated, however a number of these methods depend totally on a data-driven transformation technique which cannot fully isolate artifacts from the entire EEG signals. Also many of the exiting methods are targeted towards the removal of one or two particular type of artifacts.

In this regard, this study proposed a robust algorithm, low rank approximation based on multi-channel wiener filter that can simultaneously remove various types of artifacts (such as eye blinks, eye movement, cardiac and muscle activities among others) inherent in EEG signals, to enable accurate decoding of MI tasks in individuals with above elbow amputation. It is a semi-automatic and semi-supervised filtering algorithm for estimating a signal from noisy measurements. It has been widely applied in audio and speech related problems and used for biomedical signal processing due to its low complexity and extendibility capabilities [8], however its application to eliminate various types of artifacts inherent in upper extremity MI-EEG signals has not been investigated. Using EEGLAB to visualize the signals, the artifact removal process involves manual annotation of artifacts in the recorded EEG signals and based on the parts of the signals marked for extraction, the algorithm learns the statistical properties and subsequently apply them to automatically clean other parts of the signals that were not annotated.

Towards realizing a practical application of this method, a modified channel selection method based on sequential forward floating selection (mSFFS) was also proposed to eliminate redundant channels without affecting the classification performance.

II. MATERIALS AND METHODS

A. Participant information

A total of four transhumeral amputee subjects, all males with an average age of 41.50±7.05 years (± standard deviation) and residual limb length of 25.50±4.20 cm participated in the experiment. The objective of the study and the experimental procedure was carefully described to the participants after which they agreed to take part in the study. The experimental protocol was approved by the Institutional Review Board of Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, China.

B. Data Collection and Signal Processing

A 64-channels Al-AgCl electrodes signal acquisition system (EasyCap, Herrsching, Germany) combined with Neuroscan software was used for the signal acquisition. The electrodes were distributed over the scalp of each of the subjects based on the international 10-20 electrode placement configuration.

Preceding the placement of EEG electrode cap, subject's hair was cleaned to ensure high signal quality. Also, the electrode cap position was adjusted and the impedance between the electrode and the scalp was kept below 10 kΩ.

Before the commencement of the experiment, the subjects were trained in order to get accustomed with the experimental procedure. Each of the subjects performed five classes (a task per time) of MI tasks (which comprises of hand close (HC), hand open (HO), wrist pronation (WP), wrist supination (WS) and no movement (NM)) based on the motion image displayed through a computer screen. To avoid mental fatigue, each of the five classes of MI tasks were observed for 5s with a rest session of 5s between two consecutive tasks.

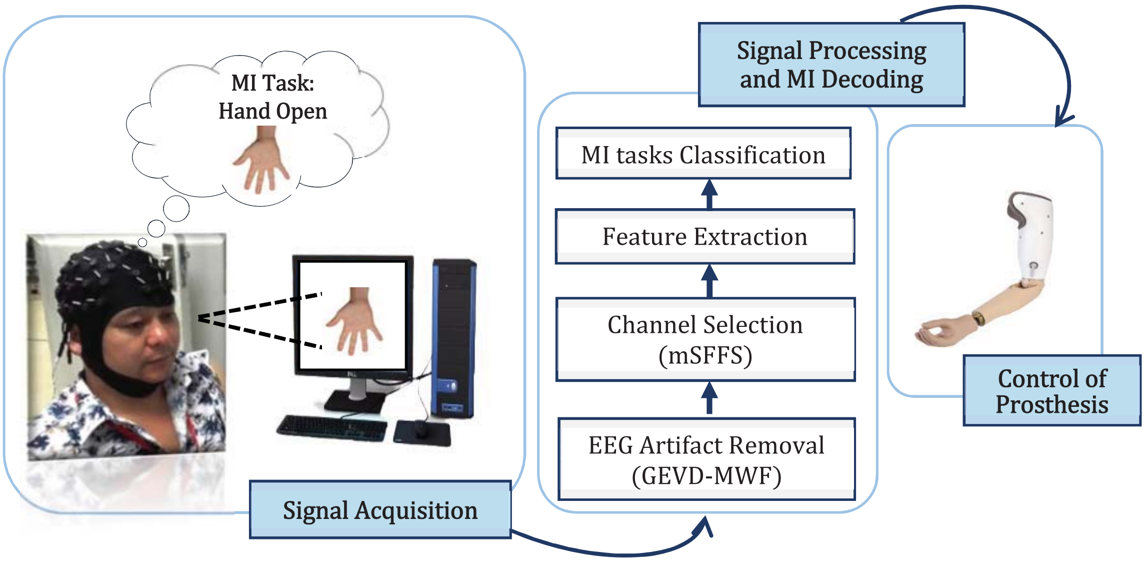

Fig 1: The conceptualized framework of the proposed control model

Five experimental trials were completed by each subject and in every trial, each MI task was repeated 10 times. In total, 250 trials per subject were collected: 5 MI tasks × 5 trials × 10 trials per MI task. It is worth noting that a pre-filter was integrated with the EEG signals acquisition system used in this study, therefore, the signal was firstly filtered with a band pass from 0.05 to 100 Hz at a sampling rate of 1000 Hz and the power-line noise was removed using 50 Hz notch filter. The recorded EEG signals were further preprocessed using MATLAB and EEGLAB toolboxes. The latter was used to visualize and extract epochs corresponding to different MI tasks. The conceptualized framework of the proposed model is presented in Fig. 1.

C. Low rank Approximation based on MWF

In this study, a robust algorithm called low rank approximation based on multi-channel Wiener filter (MWF) is proposed to process the recorded raw EEG signals. To a great extent, the algorithm can optimally remove various kinds of artifacts while minimizing the removal of the informative component of the signals. In comparison with the traditional MWF, the covariance matrix is replaced by a low-rank approximation based on generalized eigenvalue decomposition (GEVD) [8]. The algorithm descriptions is given as follows:

Suppose the recorded EEG signals, $z[t] \in P^N$ at time t with N-channels is given as

$$z[t] = x[t] + y[t] \tag{1}$$

where $x[t]$ and $y[t]$ represent the original signals and combination of different artifacts superimposed on the original signals. It is worth noting that the signals $z[t]$, $x[t]$ and $y[t]$ observed at a particular sample time $t \in M$, are treated as stochastic vectors with N-dimension and the sample index t will be ignored. The mean of the signals (signals z, x and y) are taken to be zero to satisfy the subtracting mean preprocessing step.

The covariance matrices of the signals represented as: P_{zz}, P_{xx} and P_{yy} are defined as $E\{zz\}^T$, $E\{xx\}^T$ and $E\{yy\}^T$ respectively, where $E\{.\}$ is the expected value operator. Assuming x and y are not related, then

$$P_{zz} = P_{xx} + P_{yy} \tag{2}$$

Since both P_{zz} and P_{xx} are symmetric and positive definite hence an invertible Matrix U is obtained equation (3)

$$U^T P_{zz}\, U = diag(\alpha_{z1}, \alpha_{z2}, .. ., \alpha_{zN}) = \textstyle\sum_z$$

$$U^T P_{xx}\, U = diag(\alpha_{x1}, \alpha_{x2}, .. ., \alpha_{xN}) = \textstyle\sum_x \tag{3}$$

while satisfying

$$P_{zz} U = P_{xx} U \wedge \tag{4}$$

This equation is known as generalized eigenvalues Decomposition (GEVD). The column U and the diagonal elements, $\wedge$ with an ordered matrix pair (P_{zz}, P_{xx}) corresponds to the generalized eigenvectors (GEVC) and generalized eigenvalues (GEVL), respectively
where

$$\wedge = diag(\lambda_1, \lambda_{2,} . . . , \lambda_N)$$

$$\lambda_1 = {}^{\alpha_{zi}}/_{\alpha_{xi}}, \quad \text{for i = 1,2.3, ... , N}$$

It is worth nothing that computing GEVD is the same as finding the joint diagonization of P_{zz} and P_{xx}. Therefore, with the diagonization in equation (3), we can write P_{yy} as

$$\begin{aligned} P_{yy} &= P_{zz} - P_{xx} \\ &= U^{-T} \textstyle\sum_z U^1 - U^{-T} \textstyle\sum_x U^{-1} \\ &= U^{-T} (\textstyle\sum_z - \textstyle\sum_{x)}\, U^{-1} \\ &= U^{-T} \textstyle\sum_y U^{-1} \end{aligned} \tag{5}$$

where

$\sum_y = diag(\alpha_{y1}, \alpha_{y2}, .. ., \alpha_{yN})$ and $\alpha_{yi} = \alpha_{zi} - \alpha_{xi}$

Since P_{yy} has rank Q, only the first Q diagonal elements of $\hat{\sum}_y$ will be non-zero. However, as $\hat{P}_{yy}$ generally does not have rank Q, the GEVD is used to compute a rank-Q approximation for P_{yy}. Therefore the GEVD of ($\hat{P}_{zz}$, $\hat{P}_{xx}$) and $\hat{P}_{yy}$ are calculated where the last N-Q diagonal elements of $\sum_y$ are replaced by zeros. [8, 14]. The performance of the proposed method was compared with adaptive thresholding technique [4], a commonly used approach to preprocess EEG signals.

D. mSFFS Algorithm

Generally, multichannel EEG signals contain a large number of redundant information that does not contribute to the EEG-based motor imagery classification accuracy but rather incur additional noise, high computational cost and equipment setup time. A channel selection algorithm can effectively exclude the redundant channels and select the optimal areas of brain required for accurate decoding of MI tasks [15]. Hence, this study proposed a modified sequential forward floating Selection (mSFFS) algorithm to eliminate redundant channels without affecting the classification performance.

It is an iterative algorithm that adopts a bottom-up approach to select channels with meaningful information. It removes or adds channel to an existing subset at each iteration and the selection criteria is based on maximizing the prediction accuracy. For instance, assuming C, Y_m, J(Y_m) represent the universal EEG channel sets, channel subset (which contain *m* channels) and performance accuracy of subset Y_m . Each channel in *C* goes through iteration and if a channel increases the prediction accuracy then the channel is added to the subset, Y_m , otherwise it is removed. The iteration process continue until a stopping criterion is met. In this study, the iteration stops when the current maximum accuracy (CMA) is greater than the previous maximum accuracy (PMA) and the difference is < 0.002 (CMA - PMA < 0.002). The flow chart diagram of the mSFFS is presented in Fig. 2.

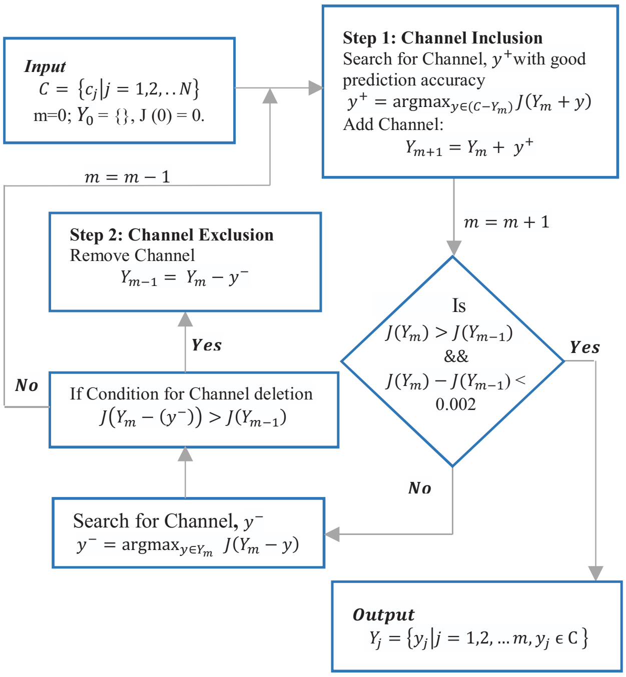

Fig 2: The process flow diagram of the modified sequential floating forward selection algorithm.

E. Feature Extraction and MI Tasks Decoding

An overlapping segmentation technique of 150ms window length and 100ms window increment was used to segment the obtained EEG signals into different analysis window. To perform feature extraction, the commonly used four time domain features, namely TD4 (mean absolute value (MAV), waveform length (WL), zero crossings (ZC), and slope sign changes (SSC)) were extracted from each analysis window. Linear discriminant analysis (LDA) classifier which has been widely utilized in EMG-PR systems due to its ease of implementation, training, and good performance was used to decode the MI tasks from the extracted features vector.

LDA algorithm was adopted because it exhibits a simple computational structure which makes it easy to implement in real life scenario [16-17]. Meanwhile, a 5-fold-cross-validation was utilized to partition the features vector into training and testing sets. Classification accuracy (CA), F1- measure, sensitivity and specificity metrics were used to evaluate the performance of the classification model.

III. RESULTS

A. Performance Evaluation of Classifiers

In reporting the results, three classifiers were constructed and their classification performances were compared. The signal for the first classifier was preprocessed with adaptive thresholding filtering method (ATSC), the second was preprocessed with the proposed method (ASC), while the third classifier was based on the selected number of channels (RCASC) from ASC. The obtained results are represented using radar plot shown in Fig. 3 and it depicts the mean classification performance of the classifiers across all the five classes of MI tasks and subjects. The arrow pointers on the plot denote the average accuracy achieved by each of the classifiers. ATSC achieved the least average classification accuracy of 77.56% against ASC (99.88%) and RCASC (96.87%). It can be deduced from this result (Fig. 3) that the various type of artifact inherent in EEG signal greatly affect the performance but with the use of the GEVD-MWF artifact removal algorithm, the performance was greatly enhanced. For practical application, the mSFFS algorithm was employed to select the most informative channels from the proposed method preprocessed signals and with 10 optimal channels selection, satisfactory average classification accuracy was achieved. Meanwhile, there is no significant difference in the classification performance of the ASC and RCASC.

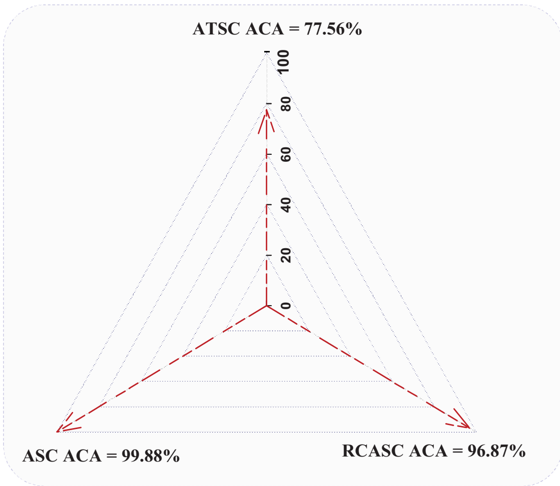

Fig. 3: Average classification accuracy (ACA) across all classes of MI tasks and subjects for ATSC, ASC and RCASC.

The performance of the classifiers was further evaluated and validated using three additional metrics (F-measure, sensitivity and specificity). Table 1 presents the metrics average performance values across all classes of MI tasks and subjects for the classifiers. These metrics basically determine how precise and robust a classifier is, and its ability in classifying the classes correctly. In comparison with ATSC, the ASC and RCASC recorded high performance scores for the three metrics. One possible reason for this is that, the proposed method significantly eliminated most of the artifacts inherent in the original signals that is affecting the classification performance. In addition, features that are highly interrelated to the classes of motor imagery task were extracted and classified. Though the ATSC achieved high specificity score (94.42%), but its sensitivity and F-measure scores simply revealed that some of the classes were incorrectly classified.

TABLE 1: THE AVERAGE PERFORMANCE EVALUATION SCORES OF CLASSIFIERS ACROSS ALL CLASSES OF MI TASKS AND SUBJECTS USING DIFFERENT METRICS

S/No.	***Evaluation Metrics***	***ATSC***	***ASC***	***RCASC***
1	***F-measure***	77.80	99.88	96.86
2	***Sensitivity (%)***	77.69	99.88	96.87
3	***Specificity (%)***	94.42	99.97	99.22

B. Classification Performance of EEG MI Tasks

The performance of the proposed method was further investigated based on individual classes of MI tasks. Fig. 4a-4c illustrate the average confusion matrix of the classification results across all subjects for ATSC, ASC and RCASC respectively. From the results, ASC recorded average classification accuracy for all the five classes of MI task that is significantly higher than ATSC. Comparing ASC and RCASC performance, it can be seen that both classifiers achieved relatively almost the same performance for the five classes of MI tasks.

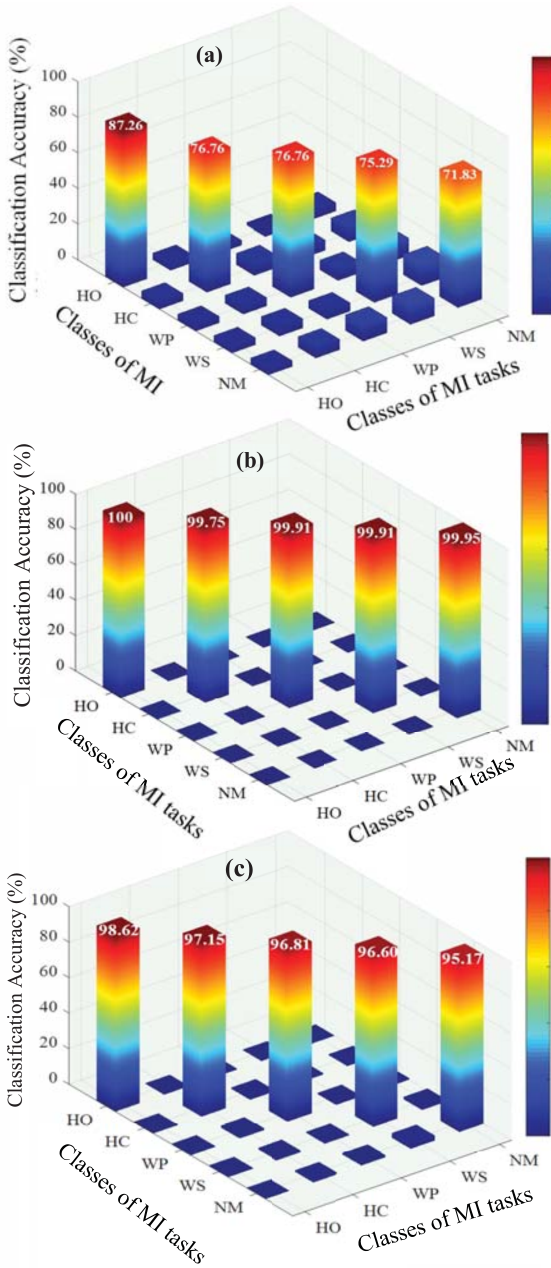

Fig. 4: Class-wise classification accuracy, average across all subjects for (a) ATSC, (b) ASC and (c) RCSC.

C. Performance Comparison of the Signal Preprocessing Methods

The effectiveness of the preprocessing methods in removing artifacts from the EEG signals was examined by selecting one representative channel, C1(27) from the 10 optimal channels and the signal plot for the proposed method and the commonly applied method are presented in Fig. 5a and 5b, respectively. The signal in blue color represent the original EEG signal while the signal in green color represent the clean EEG signal for the proposed method (5a) and common approach (5b). By analysis both plots, it could be observed that, compared with the common approach, the proposed method can effectively remove artifacts inherent in the EEG signal based on the signal reconstruction.

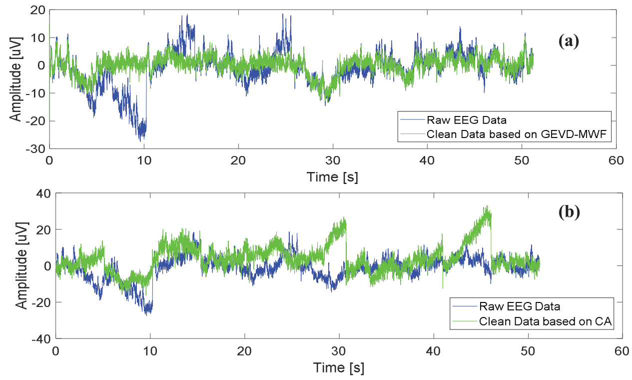

Fig. 5: Cleaned EEG signal achieved by (a) proposed method, GEVD-MWF and (b) commonly used approach for channel C1 (27). Note: CA represent common approach, the blue color signal is the raw EEG data while the signal in green color is the clean signal

IV. DISCUSSION

Due to the limited number of muscles available after amputation, individuals with high-level amputation are not able to generate sufficient EMG signals in which a number of different intended movements can be identified. Therefore, it is rare to find commercially available myoelectric pattern recognition based multifunctional prostheses for above-elbow amputees. Towards addressing this issue, multi-channel EEG signals have been considered and used as an alternative and potential method. However, EEG signals are susceptible to various types of artifacts which greatly affects the accurate decoding of motor imagery movements in amputees. In this regard, this work proposed an extendable linear statistical filtering algorithm, GEVD-MWF to remove all possible forms of artifacts contained in EEG signals recordings acquired from four transhumeral amputee subjects who were recruited for this study. Due to factors such as high computation time incurred from the 64 EEG multi-channels and equipment setup time that may affect the practical application implementation of the method, a channel selection algorithm based on mSFFS was employed to optimize the number of channels with insignificant or ignorable change in signals classification accuracy. Experimental results showed that the proposed artifact removal algorithm greatly eliminate the signal artifacts, thereby improving the classification performance.

Similarly, the channel selection algorithm was also very effective in selecting the best channels that produces high classification accuracy. More specifically, the average classification accuracy of ASC and RCASC in comparison with ATSC was improved by 22.03% and 19.31% across all the classes of MI tasks and subjects (Fig. 3). The proposed algorithms were further validated using F-measure, sensitivity and specificity metrics and it could be seen from Table 1 that they significantly outperformed ATSC across the three metrics.

In addition, the performance of the individual classes of MI tasks was evaluated and from the results presented in Fig. 4 for the three groups of classifiers. ASC and RSASC accurately classified the classes of MI tasks correctly compared with ATSC with an average range of 12.74%-27.92% and 11.36% - 23.34%, respectively. By carefully examining the results reported for ASC and RSASC, it can be concluded that there was no significant difference between there classification performance. The signal plot presented in Fig. 5a-5b further show or proof the effectiveness of the proposed method compared to the common approach.

In summary, the study results demonstrated that the proposed artifact removal algorithm, GEVD-MWF could accurately eliminate the artifacts inherent in the recorded EEG signals. Likewise, through the application of the channel selection algorithm to the artifact free EEG signals, an accuracy of 96.87% (across all classes of MI tasks and subjects) was achieved with 10 optimal channels located at the pre-motor cortex, primary motor cortex and somatosensory cortex of the brain. Finally, the outcome of this study could spur possible application of the proposed method to control multifunctional prostheses for above elbow amputee. In our future work, we hope to fully automate the proposed artifacts removal algorithm and also recruit more subjects to further validate our proposed method.

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